

The Camera, The Real-Time Loop, and How to Get a Fundus Image

RetiNova — diabetic retinopathy screening platform

Engineering and clinical-context documentation · Smart India Hackathon 2026

Every number in this document was measured on the running system, not estimated.

1. The question this document answers

Point the live camera at a wall, a hand, or a photograph of anything that is not an eye, and a screening tool should say *nothing*. If it instead returns a grade and a confidence score, that is not a cosmetic bug — it is the single most dangerous failure a screening tool can have, because a confident "*No DR*" on an image nobody could have graded sends a person away reassured.

This document explains four things, in order: what the real-time camera is actually for, what happens when there is no eye in front of it, whether an ordinary phone or laptop camera has enough resolution for any of this to be honest, and exactly what hardware and procedure are needed if you want a true fundus (retinal) image.

2. The camera does two completely different jobs

Conflating them is the source of most confusion about this project, so they are separated here first.

	Job A — Retinal imaging	Job B — Camera-only triage
What it looks at	The retina: the light-sensitive layer at the <i>back</i> of the eye.	The front of the eye and the face: pupil, iris, sclera, conjunctiva, skin.
What it finds	Microaneurysms, haemorrhages, exudates, new vessels — the actual signs of diabetic retinopathy.	Pulse rate and rhythm, pupil light reflex, conjunctival pallor — systemic risk signals, not retinopathy.
Hardware needed	A fundus lens or fundus camera is mandatory. No phone can do this unaided. Section 6 covers the options.	Nothing but the camera already in the phone or laptop.
Answers	"Does this person have referable diabetic retinopathy?"	"Of the people waiting, who should be seen first?"
In the product	Screening page → grade, heatmap, referral decision, doctor sign-off.	Health-check page → priority score, doctor's triage queue.

The physics, stated plainly. The retina sits roughly 17 mm behind the cornea and is viewed through the pupil, an aperture 2–8 mm wide. To photograph it you must put light *in* through that same small aperture and get an image back *out* of it without the illumination reflecting off the cornea straight into your sensor. Purpose-built optics solve this with coaxial illumination and a condensing lens. A bare phone camera has neither, so it images the front surface of the eye and stops there. Any product claiming retinal screening from a bare phone camera is either using an attachment it is not telling you about, or it is not doing what it says.

3. Why the analysis runs in real time, and not on a snapshot

Uploading one photograph and waiting for a verdict is simpler to build. Real-time processing was chosen because four things that matter are impossible without it.

3.1 It refuses a bad frame *before* it becomes a result

Real screening programmes reject 10–20% of images as ungradable. If that judgement only happens after the photograph is taken, uploaded, and graded, the person has usually left. Running the check on every frame means the operator is told *while the camera is still pointed at the eye*, when it costs five seconds to fix instead of a second visit.

3.2 It guides the person holding the camera, who is usually not a clinician

The intended setting is a camp, a pharmacy, or a rural clinic where the operator is a health worker, not an ophthalmic photographer. The live loop returns a target ring and one plain instruction at a time — *move closer, too bright, out of focus, centre the retina, good, hold still*. The Save button stays disabled until the frame would actually be gradable, so an unusable image cannot be submitted by accident.

3.3 It turns one attempt into best-of-many

A single frame is a gamble on blink timing, hand tremor, and a corneal reflection landing in the wrong place. Analysing a stream means the system keeps grading until a frame passes, and the prediction shown on screen is smoothed over the last five results rather than jumping on every frame. The saved image is chosen from a stream, not hoped for.

3.4 Some measurements exist only in time

This is the part a photograph cannot substitute for at all. Two of the three triage measurements are **temporal signals** — they are not visible in any still image, no matter how good:

- **Pulse (PPG).** Each heartbeat pushes blood into the skin and minutely changes how much light it absorbs. The change is far too small to see, but averaged over a region across ~20 seconds of video and band-pass filtered to the plausible heart-rate range, it is a clean waveform. One frame contains zero of this information; a video contains all of it.
- **Pupil light reflex (PLR).** The measurement *is* the change: the pupil is measured, a light is presented, and the constriction and its speed are tracked frame by frame. A sluggish or asymmetric reflex is a recognised neurological red flag, and long-standing diabetes is associated with autonomic changes in that reflex.

Only the third — **conjunctival pallor**, a proxy for anaemia — could in principle come from a still photograph, and even there the live loop earns its place by white-balancing against the sclera and using the iris as a physical ruler (≈ 11.7 mm), so the reading does not depend on the room's lighting or the phone's distance.

3.5 What it costs, measured

Frame sent for analysis	224 × 224 JPEG, quality 0.70 — the size the model consumes anyway, so nothing is wasted uploading pixels that get thrown away
Frames in flight	One. The next frame is grabbed only when the previous result returns, so a slow link lowers the frame rate instead of building a queue of stale frames
Refusing a non-eye frame	11–23 ms — the model never runs
Grading a valid retinal frame	0.6–0.9 s locally, ≈ 1.7 s on the free cloud tier
Displayed prediction	Mean of the last 5 results, so it does not flicker

The asymmetry in that table is the design. Rejection is roughly **fifty times cheaper** than grading, because the gate is deliberately classical computer vision rather than a second neural network: every number it uses is inspectable, needs no training data, and costs about a millisecond. That is what makes it affordable to run on every single frame.

4. What happens when there is no eye in front of the camera

Before any image reaches the model, four independent tests must all pass. They were chosen because a non-eye fails at least one of them decisively.

Test	Threshold	Why this catches what it catches
A lit region exists	Largest bright contour found above level 22	A black frame, a lens cap, or a switched-off camera has no illuminated region at all.
It fills the frame	$\geq 6\%$ coverage	A distant eye across a room carries too few pixels of retina to grade, however sharp.
Red dominance	$\text{mean}(R) - \text{mean}(B) \geq 10$	A retina is strongly red-dominant — 18 to 110 measured on real fundus images. Walls, skin, paper, screens and grey noise sit near zero. This one test rejects most non-eyes on its own.
Real texture	Laplacian variance ≥ 3.0	A thumb over the lens or a flat painted surface is a smooth fill; it can be red and still contain no structure. Every real photograph has texture.

If all four pass, the image is an eye, and a second set of tests asks whether it is a *gradable* eye:

Quality test	Accept range	What the user is told
Focus (green channel)	≥ 12	"Out of focus — adjust the distance slowly until vessels sharpen."
Brightness	25 – 185	"Too dark — add light" / "Too bright — reduce the light."
Contrast	≥ 12	"Low contrast — reduce glare on the lens."
Blown-out pixels	$\leq 5\%$	Over-exposure; detail in that region is unrecoverable.
Centring	within 35% of centre	"Centre the retina inside the circle."

4.1 Measured behaviour

These are actual responses from the running service, not illustrations:

Pointed at	Eye found	Quality	Grade returned	Time
Fundus image, healthy retina	yes	97 / 100	No DR	0.78 s
Fundus image, treated retinopathy	yes	90 / 100	Proliferative	0.93 s
Plain grey wall	no	0	none — refused	0.023 s
Random noise	no	0	none — refused	0.013 s
Hand over the lens	no	0	none — refused	0.018 s

A refused frame returns no grade, no confidence and no probability spread — there is nothing on screen for a person to misread as a result. It returns one sentence instead: *"No eye detected — line the lens up with the eye and hold it steady."*

If you have seen a score on a non-eye image, this is why. The gate described above is verified by an automated test suite (11 checks: three real fundus images accepted, and blurred, dark, over-exposed, noise, flat-fill and off-centre frames all refused). Any deployment that predates it will still grade a wall, because the old build had no gate at all — it sent whatever it was given straight to the model, and a softmax over five classes always sums to one, so *something* always comes back looking like an answer. The presence or absence of this gate is the difference between the two behaviours, not a change in the model.

5. Is an ordinary camera's resolution enough?

The honest answer has two halves, and the second half is the one that gets left out of most pitches.

5.1 Sensor resolution is not the binding constraint

The model consumes **224 × 224 pixels**. A 1080p video frame is downsampled to that before it is analysed, and a 12-megapixel still photograph is downsampled far harder. Every mainstream phone from the last decade already delivers more pixels than the pipeline keeps. Buying a higher-megapixel phone changes nothing about the grade.

What actually limits image quality, in order of severity: **optics** (can the light path reach the retina at all — see Section 2), **illumination** (coaxial and diffuse, versus an off-axis torch that produces glare), **focus**, and **pupil size**. All four are hardware and technique. None is megapixels.

5.2 What that resolution does cost — stated plainly

A microaneurysm, the earliest sign of diabetic retinopathy, is roughly **15–60 µm** across. A standard 45° fundus field spans about **10–12 mm** of retina. Compressed into 224 pixels, a single pixel covers roughly **50 µm** — the same order of magnitude as the entire lesion.

This is not a detail to hide; it is visible directly in the measured results. On 7,026 held-out images the model separates obviously-diseased from obviously-healthy retinas reasonably well, and performs *worst* on grade 1 (microaneurysms only) — exactly the grade whose defining lesion is at the edge of the resolution it is given. The system is therefore positioned as what the numbers support: a tool for **prioritising who a doctor sees first**, with every result reviewed by a clinician before it becomes advice.

5.3 What is measured

Evaluation set	7,026 held-out images, never seen in training
Agreement with human graders	quadratic weighted kappa 0.364
Referral decision	a threshold on $P(\text{grade} \geq 2)$, not the most likely class
At the chosen operating point	80% sensitivity, 58% specificity
What that trade means	4 in 5 people with referable disease are caught; about 4 in 10 healthy eyes are referred anyway. Deliberate: a missed case costs sight, an extra appointment costs an hour
Not assessed at all	Diabetic macular oedema — a separate cause of sight loss needing its own examination

Why report the weaker-sounding number. Taking the most likely of five grades gives a better-looking specificity (77%) but catches only 65% of referable disease — one in three people who needed a specialist would be told they were fine. A screening programme is judged on sensitivity first. Choosing the threshold deliberately moves it to 80%, and the report says on its face what that costs.

6. Getting a real fundus image: what is actually required

For the detailed retinal assessment — the grade, the heatmap, the referral decision — an image of the retina must reach the platform. There are four routes, and the platform accepts an image from any of them.

Route 1 — Existing image from any source (supported today, zero cost)

If a retinal photograph already exists — from a hospital visit, a camp, a teleophthalmology service, or a colleague's fundus camera — upload it on the screening page. JPEG or PNG. The quality gate assesses it exactly as it assesses a live frame, so an image too poor to grade is refused rather than quietly graded.

Route 2 — Smartphone + a 20 D condensing lens (₹1,500 – ₹4,000)

The classic low-cost method, well documented in the ophthalmology literature. A 20-dioptre indirect ophthalmoscopy lens plus a 3D-printed or clip-on holder that fixes the lens at the right distance from the phone.

1. **Dilate the pupil.** Tropicamide 0.5–1% drops, 15–20 minutes. *This is a prescription medicine and requires a qualified clinician* — it is contraindicated in narrow-angle glaucoma and impairs vision for several hours.
2. **Darken the room.** A dilated pupil in a dark room is the whole trick; the aperture is the constraint.
3. **Set up the light.** Phone torch on, at its lowest usable level, offset from the lens axis to keep the corneal reflection out of the image.
4. **Position.** Phone roughly 30 cm from the eye; 20 D lens about 5 cm from the eye, between phone and patient.
5. **Focus and frame.** Tap to focus on the retinal image formed by the lens, not on the lens surface. Move the lens in small increments until the optic disc and the macula are both in the field.
6. **Record video, not stills.** Then let the live gate pick the gradable frame — this is precisely the problem the real-time loop was built for.

Expect a learning curve of tens of attempts, a narrower field than a real fundus camera, and a meaningful ungradable rate. It is the cheapest route that genuinely works.

Route 3 — Purpose-built smartphone fundus adapter (₹15,000 – ₹1,50,000)

Commercial clip-on units — D-EYE, Volk iNview, oDocs visoScope and similar — package the optics, the light path and the phone alignment into one device. Substantially higher success rate than Route 2, several of them usable without dilation, and portable enough for door-to-door screening. This is the realistic sweet spot for a deployed community programme.

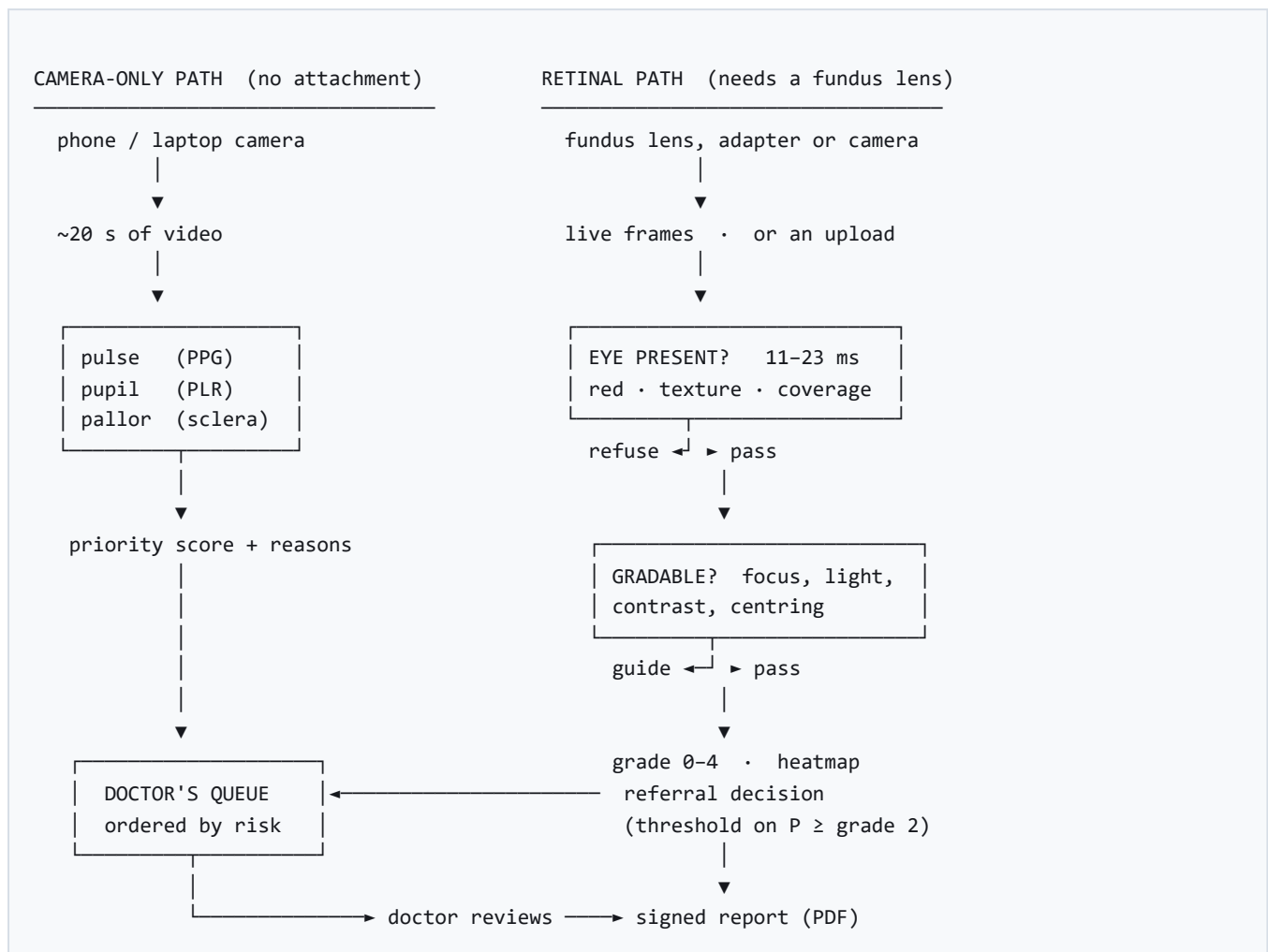
Route 4 — Dedicated non-mydriatic fundus camera (₹3,00,000 – ₹15,00,000)

Remidio Fundus-on-Phone, Forus 3nethra, Zeiss Visuscout, Topcon and equivalents: 45° field, infrared alignment, no dilation needed, consistent illumination, and the image quality the published DR-screening results are actually based on. This is what a district hospital or an established screening programme buys.

Route	Cost	Dilation	Gradable-image rate	Best fit
Existing image	—	—	as supplied	Anyone with a prior photograph
Phone + 20 D lens	₹1.5k – 4k	usually required	low, technique-dependent	Students, proof of concept, tight budgets
Clip-on adapter	₹15k – 1.5L	often not needed	moderate to good	Community and camp screening
Fundus camera	₹3L – 15L	not needed	high	Hospitals, formal programmes

The design consequence. RetiNova does not require any one of these. The camera-only triage path (Job B) runs on hardware every patient already carries, and gives a doctor a defensible ordering of who to see first. The retinal path (Job A) turns on the moment any of Routes 1–4 supplies an image. The two are independent, so the platform is useful before a single rupee is spent on optics, and better with each step up.

7. How it fits together



8. Summary

- **Real time is not a demo feature.** It is what rejects unusable frames while the eye is still in front of the camera, what guides a non-specialist operator, and what makes pulse and pupil-reflex measurable at all — none of which a snapshot can do.
- **A non-eye returns nothing.** Four cheap tests refuse it in about 15 ms, before the model runs. Silence is the correct output, and it is the safety property that matters most.
- **Resolution is not the limit; optics are.** The model sees 224×224 pixels regardless of the sensor. The real limits are the light path and the pupil — and the cost of that resolution is visible and reported, not hidden.
- **A fundus image needs a lens, at any budget from ₹1,500 to ₹15 lakh.** The platform accepts an image from all four routes, and works usefully before any of them is bought.
- **Every result is reviewed by a doctor.** The AI orders the queue; the clinician decides.

RetiNova is a screening-support prototype. It is not a diagnostic device, it is not regulator-approved, and no output should be acted on without clinician review. Model performance figures are measured on a public research dataset and would need prospective validation on Indian screening populations before any clinical deployment. Dilating drops are prescription medicines and must be administered by a qualified clinician.